

On "Dead Times" and Free Thinking

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I no longer have "dead times." Every scrap of time that slips free of the attention-demanding tasks of everyday life — like waiting for an elevator, or the walk to the car — instantly becomes, for me, catch-up time. Mail, WhatsApp, Twitter, Facebook. So that nothing goes to waste.

This sounds reasonable, except that despite my instinct to make use of every free, wasted moment for productive activity, a feeling has built up in me of — how to put it correctly? — a kind of suffocation. And I paused to look a little at what happens to me in these "dead" times, when I do not immediately channel them toward some defined, useful mental activity.

In what follows I will argue that this compulsion to "inject content" into content-empty moments leads us somewhere not so good. And, practically, I will suggest that it is worth our while — mainly from a cognitive standpoint — to leave as many "dead" times as we can in which we do not look at our phones. Does this sound like a cliché? Well, perhaps — there are some wise clichés going around the world too — but I want to propose a broader conceptual foundation for the discussion, one that deals with the different kinds of thinking that can occupy our consciousness. Mainly, I will try to put my finger on, and to define at length, what I felt was missing when I fill the gaps in my time with social activity or content-consumption — the opportunity for my thoughts simply to wander.

Reactive Thinking

In what follows I will distinguish between two kinds of thinking — two forms of mental-cognitive activity that differ from one another in several important respects. The first I will call reactive thinking, and, opposite it, free thinking. In brief, reactive thinking is thinking within a practical or social context, directed by stimuli or by goals. Free thinking, by contrast, is what I mean by our mental activity when we are not within such a context — to illustrate, when we are walking down the street, tying our shoelaces, eating breakfast (alone), driving, or waiting for the microwave. I call it "free" to describe its wandering, independent character. But in order truly to justify the label "free thinking," I will first need to clarify what characterizes its negative image, reactive thinking. For this purpose, I will divide reactive thinking itself into two kinds, which I will call active reactive thinking (goal-directed) and passive reactive thinking (stimulus-directed).

Active thinking is what we usually call "thinking" — an effort to solve some problem set before us, the active and conscious mental search for a fitting course of action — usually a response to the circumstances in which we find ourselves. Active thinking directs attention toward some goal, something we want to achieve, and works out and investigates for us the way to achieve that goal. But this is usually not a matter of higher-order goals like how to advance

professionally or find more satisfaction in one's relationship. Goals accompany our everyday life at every step, in every place where we have some desire and seek to realize it in reality. For example, when we need to get to a meeting room and are weighing which is the best route for us; when we need to decide what to order from the café menu so as to maximize the cost-to-taste-to-fullness ratio; and even when a colleague walks into the room and says something to us, and we need to plan out, within a few tenths of a second, how we will respond. Naturally, higher levels of thinking and planning too — the kind required when we perform a complex, non-routine action, like cooking stuffed zucchini or preparing a presentation for a talk — are included within active thinking.

Passive thinking, by contrast, is what occurs in our consciousness in response to sensory stimuli of any kind. When we see something that catches our attention, hear someone speaking, read advertising signs, or listen to a song on the radio, the sensory signal is processed in our brain and activates all the layers of our consciousness in all sorts of directions. If, for example, I am listening to a song, my passive reactive thinking might be a feeling that rises up in me, a musing on the meaning of the words, remembering another song that resembles it, and so on. All of these, in turn, arouse chain reactions of further musings, feelings, analogies and other sensations, until I move on to occupy myself with something else, or, alternatively, am exposed to another sensory stimulus that now "pulls" my attention toward it and, around it, reawakens my system of passive thinking anew. The first circle of these mental responses — the thinking that arises immediately in the face of an external stimulus, without any willed or inner direction from consciousness — is what is called here passive thinking.

Passive and active reactive thinking trade shifts with one another at a very rapid pace. Let us consider, for example, what happens while I am opening WhatsApp. I read and process the new messages in one of the groups (passive), and almost immediately think about how to compose a reply — ideally empathetic, useful or amusing, depending on the context (active). When I finish writing, I check to see whether someone is currently typing a reply, and if so this makes me curious and I try to guess what they are writing (passive), and, if they do write it, what I will answer them (active). Alternatively, if no one is typing at the moment, I wonder whether it's worth opening a news app or not (active), and from there it goes around again. That is, the decision I make (through active thinking) will determine which stimuli I am exposed to; these will trigger mental response processes in me (passive thinking), and, joining with my own desires, will also lead to the selection of new goals and to active thinking for the sake of decision-making.

The Freedom to Not React — Free Thinking

These two forms of thinking — active and passive — seem so different from one another that it is presumably hard to understand why I lump them together under a single overarching category of reactive thinking. But I believe they share a significant common denominator. What characterizes reactive thinking as a whole is that the focal point around which the thinking revolves is set externally, for consciousness.

In passive thinking this is especially prominent, since this kind of thinking can be described as an instinctive response to stimuli that changes constantly in the wake of changing stimuli — like a bouncy ball tied by a string to a moving car. The path of passive thinking has a considerable portion that is original and its own, but it is constantly chasing after a rapidly changing environment.

Active, planning-oriented reactive thinking, too, in fact walks along paths that are set for it in advance. Here, it is true, the external focal point that leads the thinking is not a sensory stimulus but a goal. Goals, which are both the fuel and the GPS of active thinking, are drastically shaped by circumstance; in every kind of reality, in every environmental state, my will fixes upon a new immediate target and directs the tools of thought toward it, which will analyze the situation and supply the "how" instructions. But I claim something more than this — it is true that goals depend not only on circumstance but also on internal factors, that is, on our own desires. And yet, even so, this is not a natural flow of our thinking, arising from its own branching, wandering nature, but rather thinking that is channeled, led by some higher-order consciousness — such as my will to solve a problem and to invest directed cognitive effort in solving it. So this is not necessarily thinking that "darts about" the way passive reactive thinking does, but rather orderly thinking that has organizing principles. But here, too, the thinking does not simply "go its own way," at a pace and along a path inherent to the thinking itself, but rather advances while harnessed to a force external to it (the will). In this sense, both passive thinking and active thinking are responses of the cognitive system to an experience, to something that is not, in itself, part of the cognitive sphere — whether these are real stimuli arriving from the senses, or practical goals and targets arising from the psychic forces that drive us.

From what has been said so far, a picture already begins to emerge of what I have called "free thinking." By this term I mean any mental-cognitive process that is not directed toward a stimulus or a goal, that does not operate as a response to an external source of experience. Most of the time, we don't really count these cognitive processes as "thinking"; some of them we barely notice, and some we don't "see" at all. And yet, these mental processing procedures deal with images and sensory experiences, with concrete and abstract concepts and ideas, with memories and plans — in short, with all the domains in which we deliberately and consciously handle what we would all agree is "thinking."

As noted, these are thought processes that occur within us during "dead times" — that is, when "we don't need to think about anything." It seems there is no real conscious state (waking, at least) of non-thinking; the cognitive system works almost all the time, much like our other bodily systems, such as the hormonal or immune systems. When we do not willfully direct our thinking toward something particular (and prevent our sensory environment from directing it), it keeps working, in the background or not, on "its own material." These are times when we process experiences, replay episodes and re-imagine them, analyze a concept we've encountered, bring two ideas we've heard into contact with one another, leap from association to association, from idea to image, from picture to concept. And we are not necessarily aware that we are doing any of this.

The Cognitive Unconscious

Free thinking, or "mind-wandering," is a phenomenon that exists across a range of different levels of awareness. It can be fully exposed to our awareness, such that we even feel we are controlling the stream of our thought to some degree or another; it can happen in a kind of "inattention," at a low level of focus and alertness to the process; and it can even be entirely unconscious to us, so far below the radar that we could never report on this thought process at all.

The idea that there could be "thinking" that is not conscious sounds, at first glance, strange and counterintuitive. Nonetheless, the scientific literature in the field of cognition embraces this concept, which has received confirmation from all sorts of empirical experiments in recent decades, given that, as noted above, the term "thinking" refers to a much broader range of mental information-processing procedures than what we are used to calling "thinking" in everyday language. Any such processing procedure that scientists manage to capture by way of their experiments, and which nonetheless does not occur in an explicit, declared way within the conscious awareness of the experimental subjects, will be labeled unconscious thinking — part of "the cognitive unconscious," or the unconscious mind.

In cognitive-psychological research, the most common example of an unconscious thought process is called priming. This is a general name for cases in which exposure to a certain stimulus affects the way we perceive and process a related or similar stimulus immediately afterward. Instead of describing an experiment, let me illustrate this with a sequence of riddles a nephew of mine once played on me —

"What color is a polar bear?"

"My flag is blue and..."

"What does a cow drink?"

The purpose of this trick is to get the "victim" to answer, mistakenly, "milk" — and indeed, this is the answer that leaps to mind immediately. The reason for this is priming — since you were occupied with the concept "white," the associative network in your head sent you (following "drink" and "cow") first of all to the beverage associated, for you, with that color, even though, had you encountered the last question on its own, you would probably have thought only of water. This is a demonstration of semantic priming. In everyday life, we experience priming, for example, when, after buying a car of a certain model, we suddenly notice that there's a car like it on every corner; since that car is present in our background thoughts, our consciousness selects, from among the sensory stimuli we are exposed to, and highlights, those connected to it. Research literature shows that the priming effect exists even when the initial stimulus (for example, the word "white") is presented subliminally (for example, a very brief exposure), meaning it is not registered in conscious thought but only in an unconscious sensory way. But even when the stimulus is conscious, the thought process that "activates" and highlights all the

concepts related to "white" (and then the concepts related to those, and so on) is not itself conscious; it is a processing procedure of the unconscious mind.

But unconscious thinking also includes processing procedures far more complex than associations, and empirical experiments have shown it to have real importance even for high-level tasks like problem-solving. Studies have shown that when people are presented with a difficult problem that requires resourcefulness and creativity, a period of time in which they (willingly) stop thinking about it contributes to their ability to solve it. This phenomenon is called "the incubation effect." The basic experiment goes like this: the same problem is given to two groups of people — the first group is asked to work on the problem continuously for about two hours; the second group is given an hour to work on the problem, and then made to engage in a different activity (for example, other tasks) for a period of time called the incubation period. Afterward, they are given another hour to work on the original task. Many experiments, in various versions, have shown the same effect — the second group succeeds better at solving the problem, and succeeds more the longer the incubation period is and the lower its cognitive load.

What explains such an effect? The basic idea is that we can have several parallel lines of thought at once, some more in the foreground of awareness and some less so. While conscious thinking has moved on to occupy itself with something else, some thread of thought concerning the problem develops in a more associative, unrestrained and spontaneous way, and gives rise to new, creative directions of thinking that lead to the solution. It is precisely the interruption of continuous, focused engagement with solving the problem that allows for the emergence of "intuition" and of "outside the box" insight, which advances our problem-solving on several levels. In this context, it doesn't much matter how consciously or unconsciously this thinking occurs; the point is that these are threads of thought that develop and unfold not out of willed intention and direction, but spontaneously, by the very nature of consciousness to think.

Incidentally, in everyday life many people are familiar with the incubation effect from cases in which they went to sleep troubled by some hard, challenging problem and woke up in the morning with a kind of "illumination" pointing to a better way of tackling it. We think even in sleep — perhaps the freest thinking there can be.

A Timeout from the Here and Now

Free thinking, then, denotes thought processes that arise and are driven by an inner cognitive engine (our own consciousness). It is not chained to the sensory stimuli we encounter, and does not carry the immediacy and urgency of the practical tasks of some given present moment. We have therefore characterized it as a kind of release from the "here and now" — which is, in fact, one of the most basic human mental capacities: what allows us, as human beings, to create and develop independent, abstract cognitive content unconnected to what is happening, in the present, in our current surroundings. Empirically, "mind-wandering" has been found to be linked to a detachment from the present, in that it contributes to the advancement of long-term personal goals and aims — such as, for example, the understanding that the value of some

desirable thing does not diminish merely because it lies far in the future (an understanding whose opposite is called delay discounting bias). The mechanism of mind-wandering disconnects our attention from what is happening around us, and redirects focus from processes of perception and response toward more mental, abstract content — a kind of zoom-out of consciousness away from the arena of present reality.

In this sense, it seems that free thinking is a resource worth protecting. Our unintentional thoughts take us to all sorts of places that lie beyond the ongoing present, and, on the whole, these seem to be places that develop us — whether these are the seeds of creative, original ideas; critiques of beliefs and conceptions held by us or by those around us, and their confrontation with new data and arguments; the consolidation of a general memory out of an accumulated collection of impressions; introspective self-understanding through the replaying of feelings and memories; the clarifying and refining of aspirations and desires that nest within us and seek their way to fulfillment; and more. Consciousness rolls toward these thoughts naturally, out of our psychic and cognitive needs; but a necessary condition for this is stretches of time in which it is free from chasing after stimuli or goals.

The Social Environment — A Source of Response-Triggering Stimuli

When do we think freely, and when reactively?

The answer to this is bound up with the question of which contexts demand a great deal of our attention and which demand less. And here I return to the definition of "dead times" with which I opened, and to why consuming content or engaging in social interaction through media, in those very moments, "suffocates" them. My main claim is that any social context immediately draws us into reactive thinking. Social environments are environments that demand a great deal of cognitive energy, and they occupy the whole of our attention. Society — and, in fact, any other person — is an endless source of unpredictable stimuli to which I will need to respond in new ways that require planning. To illustrate the claim, let us characterize the negative image of these social contexts — circumstances in which our reactive thinking is not intensively engaged, allowing thought to wander freely.

Free thinking arises mainly when we are engaged in routine, repetitive activity. This is activity we engage in alone, that deals with things rather than people, that is mainly motor in nature, and that we are accustomed to and in control of. Take driving, for example. Even though this is a fairly complex activity, requiring alertness to many elements in the world around us and careful planning of our smallest movements, we feel that our consciousness is, in fact, fairly free during these minutes of driving, and we find ourselves musing on all sorts of things, or absorbed in conversation, or listening to the radio. This is made possible because the "rules of the game" of driving remain fairly constant; the "environment" in which the activity of driving takes place is one whose behavior we are very used to. What this allows is for the sensorimotor response system responsible for driving to enter an "automatic mode" and free up resources for other mental activity taking place at the same time.

In biological terms, a complex but repetitive motor skill (like driving) becomes, after we practice it for a period of time, a simple motor activity, for which our brain has already acquired a kind of fixed "neural pathway" that operates it. The brain therefore transfers this knowledge to a lower part of itself — the cerebellum, which is responsible, among other things, for motor coordination — and frees up the higher parts (the cerebral cortex) for more conscious cognitive activity. The cerebellum is considered "lower" because, evolutionarily, it is more ancient, and is structurally shared by many, less-developed animal species. I think it would therefore be justified to describe it as part of the "pre-social" part of the human brain — the part that knows how to handle a predictable environment governed by rigid rules (the laws of physics).

In contrast to these contexts of routine activity, social contexts are, generally, environments containing far more unpredictable stimuli that seize our attention, and in which there is a constant need for responses fitted to a dynamic situation. As a result, they demand more attention and alertness from us. I think this is also why a setting of social interaction feels like a more "alive" environment, one in which we feel more vital — because it truly activates our senses and puts us into a state of alert, energetic attention. This is also one of the sources of our deep need for social life, at almost any time whatsoever, and of our pull toward the Facebook app on the device in our pocket.

The Smartphone Reflex

What would happen if we eliminated all our dead times? What would happen if we plugged up every gap our thoughts have to wander in, and filled them instead with some other social stimulus — if we "made use" of every free ten, or twenty, or two hundred seconds we have while taking out the trash or walking alone down the street, to glance at our phone and catch up on whatever needs catching up on? This reflex — in which I open up the world of information and socializing on my smartphone at every moment of cognitive release — feels, to me, to be becoming more and more common. It is an entirely logical behavior, since it feels more right to us, more "alive" and more enjoyable to take part in a virtual social interaction (or to consume current information, which is also an indirect kind of social interaction) than to "do nothing." But is this an adaptive behavior? Is something fundamental to the experience of existing, and to our mental-cognitive mechanisms, not going to be lost?

I have no answer to these questions. I also believe one should be wary of the conservative instinct that sees, in every social-technological change, the end of humanity as we have known it, since doomsday prophecies are usually proven wrong, and humanity adapts itself to new frameworks and tools. On the other hand, I also think that we live in a reckless culture that respects human habits at almost no level whatsoever, and that we are very far from giving thought to all the aspects and implications of the changes that are altering all our lives beyond recognition.

So, as I said, I do not know what the long-term consequences of the smartphone reflex might be for our life, our psyche and our consciousness. But my sense is that this is not a path of healthy

development. My feeling is that our consciousness is not accustomed to having its windshield sealed off with stimuli at every given moment. Human beings need this time, in which thinking is freed from any constraint or stimulus, in which one can muse on general thoughts, bring up memories, examine feelings, reconsider things, cast doubt, find support for new or old ideas, look again at familiar things, and simply let thought roll along on its own. This is oxygen for our emotional processing, for creativity, for introspection. In my view, free thinking is cleaner and more authentic than any interpersonal interaction; it contributes greatly to our psychic balance and to our depth of thought and of existence, and it deserves to be given the space and the respect it is due.